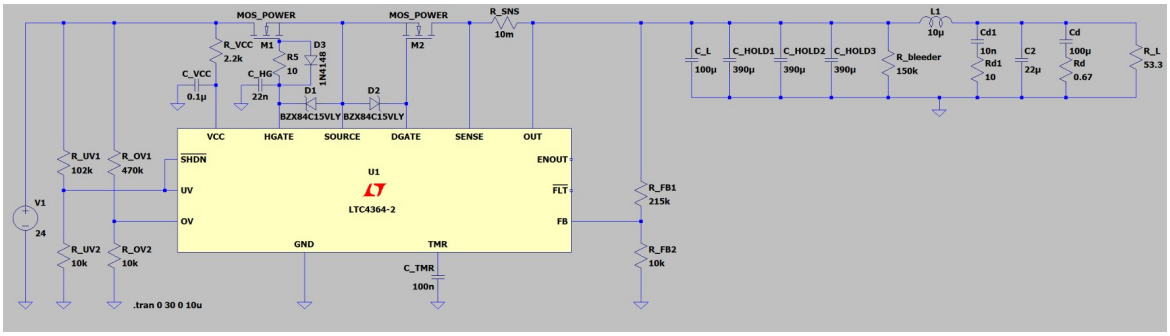
DOC-ANALOG-007

1 Executive Summary and Input Bus Specifications

The actual Issue establishes the primary power conditioning and active protection stage for the Analog Computer for Control Dynamics. The module receives an unregulated supply from a nominal 24 V_{DC} industrial bus and delivers a clean, damped, and surge-protected supply rail (V_{buck_in}) to the downstream secondary Buck converter and precision bipolar linear regulators (±15 V).



Physical Operating Requirements

- *Nominal Supply Voltage (V_{nom}):* +24 V_{DC} (18 V to 32 V continuous operating window).
- *Active Surge Clamping:* Clamped to V_{clamp} = 28.125 V during +80 V input surges (Load Dump transients spanning 15 ms to 100 ms).
- *Reverse Polarity Protection:* Absolute blocking with zero leakage current under bus collapse down to -30 V.
- *Brownout Ride-Through (t_{hold}):* Minimum hold-up time t_{hold} ≥ 13.6 ms above 18 V under brownout to 0 V at 10.8 W nominal load.
- *Inrush Current Limitation:* I_{inrush} ≤ 550 mA via a controlled *dV/dt* Miller ramp into a 1.17 mF capacitor bank.
- *Input Loop Stability:* Compliance with Middlebrook Stability Criterion ($|Z_{out,filter}| \ll |R_{in,buck}|$).

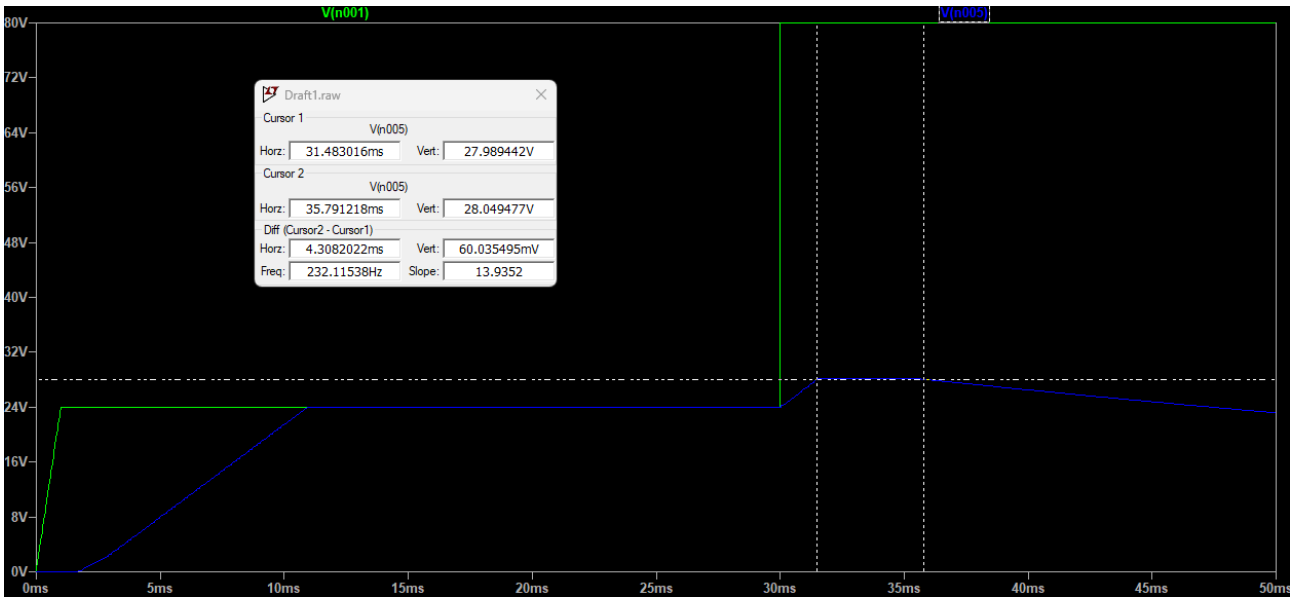
2 Monolithic Active Architecture (LTC4364-2) and Back-to-Back Power Stage

To eliminate forward conduction power losses associated with conventional Schottky diodes ($V_f \approx$ 0.7 V) and avoid the progressive degradation of Metal Oxide Varistors (MOVs), the design utilizes an LTC4364-2 surge stopper with an integrated ideal diode controller. The pass element comprises two discrete N-Channel power MOSFETs (MOS_POWER) connected in a common-source (back-to-back) configuration.

2.1 Linear Overvoltage Regulation (M_1)

During an input surge (V_{IN} > 28.1 V), the internal voltage amplifier (*VA*) controls the HGATE pin, modulating M_1's gate-to-source voltage V_{GS} to operate the MOSFET in its linear/active region. The feedback divider connected to the FB pin sets the maximum clamped output voltage:

$$V_{clamp} = V_{th,FB} \cdot \left(1 + \frac{R_{FB1}}{R_{FB2}}\right) = 1.25\ \text{V} \cdot \left(1 + \frac{215\ \text{k}\Omega}{10\ \text{k}\Omega}\right) = \mathbf{28.125}\ \text{V}$$



3 Physics of the SOA Fault Timer (C_{TMR}) and Auto-Retry Cooling Dynamics

During the +80 V surge event, transistor M_1 drops $V_{DS} \approx 51.9$ V across its channel while conducting full load current. To prevent junction temperature T_J from exceeding the Safe Operating Area (SOA) energy limit (P^2t), the LTC4364-2 charges the TMR pin with a current proportional to $V_{CC} - V_{OUT}$:

$$I_{TMR(UP)OV} = 2\,\mu\text{A} + 0.644\left[\frac{\mu\text{A}}{\text{V}}\right] \cdot (V_{CC} - V_{OUT} - 0.5\text{ V})$$

For V_{CC} - V_{OUT} = 51.9 V, the pull-up current equals $I_{TMR} \approx 35.1$ µA. With C_{TMR} = 100 nF, the maximum allowable linear regulation period before fault trip is:

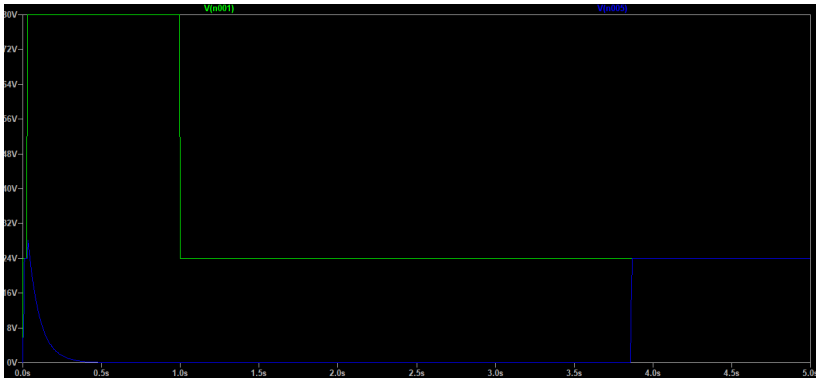
$$t_{OV} = \frac{C_{TMR} \cdot 1.25\text{ V}}{I_{TMR}} + \frac{C_{TMR} \cdot 100\text{ mV}}{5\,\mu\text{A}} \approx \mathbf{5.5}\text{ ms}$$

3.1 Auto-Retry Cooling Cycle (LTC4364-2)

When V_{TMR} reaches 1.35 V, M_1 turns off and the controller initiates 32 charge/discharge cycles on C_{TMR} between 0.15 V and 1.35 V at 2 µA:

$$t_{cool} = \frac{63 \cdot C_{TMR} \cdot 1.2\text{ V}}{2\,\mu\text{A}} = \frac{63 \cdot 100\text{ nF} \cdot 1.2\text{ V}}{2\,\mu\text{A}} = \mathbf{3.78}\text{ s}$$

The resulting thermal duty cycle D = 5.5 ms / 3780 ms ≈ 0.145% guarantees complete silicon thermal relaxation prior to auto-retry.

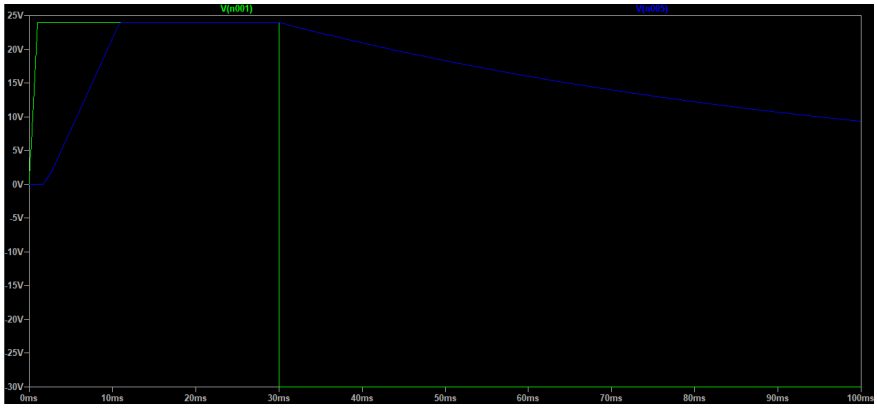


4 Reverse Polarity Protection Dynamics (-30V)

Transistor M_2 operates as an ideal diode. The controller monitors the differential forward drop between the common SOURCE node and the SENSE pin (ΔV_{SD}).

- *Forward Conduction:* The DGATE pin is pumped above SOURCE to regulate a forward drop of ΔV_{SD} = 30 mV across M_2, reducing static power dissipation ($P = I^2 \cdot R_{DS(on)}$).
- *Reverse Blocking:* When V_{IN} collapses to -30 V, current flow reverses. As soon as ΔV_{SD} < -30 mV, an active fast pull-down circuit discharges DGATE to SOURCE with a 130 mA sink current in t < 0.35 µs.

When M_2 opens, its intrinsic P-N body diode becomes strongly reverse-biased relative to the load, preventing reverse discharge currents from C_{HOLD} back into the faulted input bus.



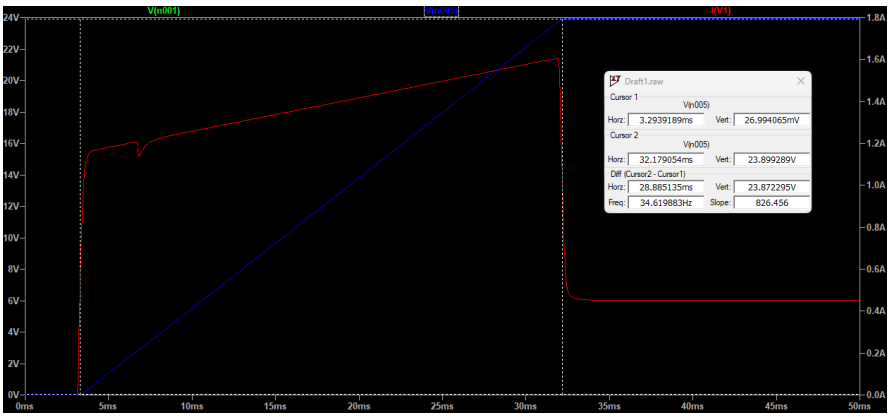
5 Inrush Current Control and Hold-Up Capacitor Bank (C_{HOLD})

5.1 Miller Gate Ramp (C_{HG})

To prevent destructive inrush current spikes into the 1.17 mF bank during power-up, a Miller capacitor C_{HG} = 22 nF connected to HGATE limits the output voltage slew rate:

$$\frac{dV_{out}}{dt} = \frac{I_{HGATE(UP)}}{C_{HG}} = \frac{20\,\mu\text{A}}{22\text{ nF}} \approx 0.909\text{ V/ms} \quad (909\text{ V/s})$$

$$I_{inrush} = C_{total} \cdot \frac{dV_{out}}{dt} = 1.292\text{ mF} \cdot 0.4545\text{ V/ms} \approx \mathbf{587}\text{ mA}$$

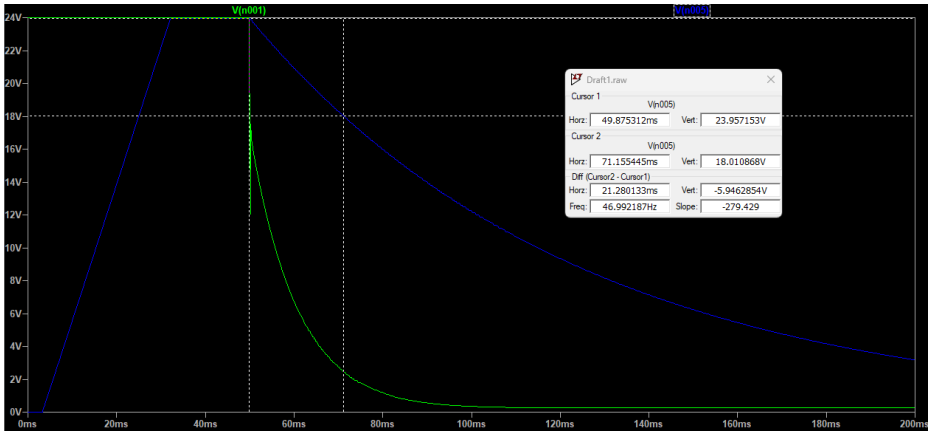


5.2 Hold-Up Capacitance Sizing

During a total input loss (V_{IN} to 0 V), the energy required to support 10.8 W of constant-power load desynchronizes from V_{nom} = 24 V down to V_{min} = 18 V:

$$C_{HOLD} = \frac{2 \cdot P_{load} \cdot t_{hold}}{V_{nom}^2 - V_{min}^2} = \frac{2 \cdot 10.8\ \text{W} \cdot 13.6\ \text{ms}}{24^2 - 18^2} = \mathbf{1166\,\mu F}$$

The bank is implemented using 3 parallel polymer hybrid aluminum capacitors of 390 µF/35 V ($C_{HOLD,total}$ = 1170 µF, *ESR* < 7 mΩ).



5.3 Safe Passive Discharge ($R_{bleeder}$)

A passive bleeder resistor $R_{bleeder}$ = 150 kΩ is placed in parallel with C_{HOLD} to discharge residual energy upon main system de-energization:

$$t_{discharge} = R_{bleeder} \cdot 3 \cdot C_{HOLD} = 150\ \text{k}\Omega \cdot 3 \cdot 1.29\ \text{mF} \approx \mathbf{571.5}\ \text{s} \approx 9.5\ min$$

The value of $R_{bleeder}$ prioritizes efficiency and steady-state PCB thermal stability (P_{loss} ≈ 3.84 mW), preventing thermal gradients that would induce offset drifts in the precision operational amplifiers of the analog core.

6 Gate Oxide Immunity (V_{GS}) and Primary TVS Protection

6.1 Gate Oxide Protection Network

1. *Zener Clamp Diodes (*D_1, D_2*):* Two 15 V Zener diodes (BZX84C15VLY) connected across GATE and SOURCE of M_1 and M_2 prevent high *dV/dt* transients or capacitive feedthrough from exceeding the absolute gate-oxide breakdown rating ($V_{GS,max}$ = ±20 V).
2. *Fast Gate Discharge Diode (*D_3*):* A 1N4148W diode is connected in antiparalel across damping resistor R_5 = 10 Ω. During short-circuit events, D_3 conducts in forward bias, allowing the 130 mA fast pull-down driver to bypass R_5 and rapidly discharge input capacitance C_{iss}.

6.2 Primary Input TVS

A unidirectional TVS diode SMAJ70A installed directly at input node N001 clips ESD and sub-microsecond inductive surges to below 70 V before the LTC4364-2 active loop responds.

7 Output π-Filter and Middlebrook Stability Criterion

The downstream Buck DC-DC stage exhibits a negative incremental input resistance $R_{in,buck} \approx -53.3\ \Omega$. To prevent relaxation oscillations on the 24 V bus, the output π-filter (L_1 - C_2) includes a parallel damping network (C_d - R_d).

7.1 Middlebrook Damping Equations

1. *Critical Damping Capacitance (*C_d*):*

$$C_d \geq 4 \cdot C_2 = 4 \cdot 22\,\mu\text{F} = \mathbf{88\,\mu F} \quad \longrightarrow \quad \text{Selected: } 100\,\mu\text{F}$$

2. *Damping Resistance (*R_d*):*

$$R_d = \sqrt{\frac{L_1}{C_2}} = \sqrt{\frac{10\,\mu\text{H}}{22\,\mu\text{F}}} \approx \mathbf{0.67\,\Omega}$$

The filter's peak output impedance is clamped to $|Z_{out,filter}|_{max} \approx R_d = 0.67\ \Omega$. Since $0.67\ \Omega \ll |-53.3\ \Omega|$, the control loop phase margin of the Buck regulator remains intact (> 38 dB stability margin).

7.2 High-Frequency Snubber Decoupling ($C_{snub} - R_{snub}$)

To absorb high-frequency L_1 *di/dt* inductive spikes without degrading the Middlebrook low-frequency damping, a parallel snubber branch ($C_{snub} = 10$ nF, $R_{snub} = 10\Omega$) is added:

- *At Filter Resonance (*$f_0 \approx 10.7$ kHz*):* Snubber impedance $|Z_{snub|} \approx 1483\ \Omega \gg |Z_{Cd\text{-}Rd|} \approx 0.686\ \Omega$. Resonance damping current flows exclusively through $C_d - R_d$.
- *At High Frequency (*$f \geq 10$ MHz*):* Equivalent Series Inductance (ESL) blocks C_d, while C_{snub} acts as a low-impedance path, enabling $R_{snub} = 10\ \Omega$ to disipate *di/dt* switching spikes.